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The Association between cytokine levels, sepsis severity, and clinical outcomes in sepsis: a quantitative systematic review protocol.

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Title: The Association between cytokine levels, sepsis severity, and clinical outcomes in sepsis: a quantitative systematic review protocol.

Abstract:

Introduction: It is widely assumed that **s**epsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection mediated by an increase in multiple pro-inflammatory cytokines. The levels of key pro-inflammatory cytokines tumor necrosis factor (TNF), interleukin-1 (IL-1)β, and interferon (IFN)γ are poorly characterized during sepsis. We believe this project will produce a "gold standard" document to which other reports on cytokine levels will be compared. The objective of this systematic review will be to identify key cytokines circulating levels in patients with sepsis and assess the association between these levels and the morbidity and mortality outcomes related to sepsis.

Methods and Analysis: We would include reports of any design except for case reports. Sepsis patients will comprise those with a diagnosis of sepsis, severe sepsis, or septic shock. The primary exposure is levels of three pro-inflammatory cytokines. The primary outcome is mortality at 28 or 30 days. Study subjects can be of any age, sex- or ethnicity. Studies will be restricted to the English language. Medline, Embase, Cochrane Library, and Web of Science Core Collection will be searched for eligible studies. A database search will include studies from 1985 to May 2020. Two reviewers will independently screen and select studies, assess methodological quality, and extract data. A meta-analysis will be performed, if possible, and the Grading of Recommendations Assessment Development and Evaluation (GRADE) Summary of Findings presented.

Ethics and Dissemination Formal ethical approval is not required as data will be extracted from existing literature. This systematic review will be disseminated through a peer-reviewed publication and at conference meetings.

PROSPERO Registration Number: CRD42020179800

Keywords: sepsis; cytokine storm; IFN-γ; IL-1β; TNF.

Word Count: 3202

Article Summary

Strengths and Limitations of this study

- This study will systematically assess the levels of key cytokines in sepsis patients and study the relationship between these levels and sepsis outcomes such as organ dysfunction and mortality.
- A systematic review of this topic will contribute to a better understanding of the pathophysiology of sepsis and inform therapeutic approaches to anti-cytokine therapy in patients with sepsis.

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- A possible publication bias might limit this study as we will include only peer-reviewed published data. Therefore, this systematic review might not capture evidence from other sources.

Introduction:

Sepsis is an important global health problem associated with significant morbidity and mortality. In 2017, there were an estimated 48.9 million cases of sepsis and 11 million sepsis-related deaths. Approximately 19.7% of all deaths recorded in 2017 were attributed to sepsis.[1] In the United States (US) in 2013, the annual cost of sepsis was estimated at $23.7 billion, representing the largest expenditure for any disease treated in US hospitals.[2] In 1992, sepsis was defined as a systemic inflammatory response syndrome in the presence of a documented or clinically suspected infection.[3] In 2016, sepsis-3 diagnostic criteria were introduced, and sepsis was defined as "life-threatening organ dysfunction due to a dysregulated host response to infection." SEPSIS-3 defined SEPSIS as an increase in sequential organ failure assessment (SOFA) score by two or more points. Before SEPSIS-3, severe sepsis was defined as sepsis with associated dysfunction of one or more organ systems. Septic shock is a subset of severe sepsis in which hypotension refractory to fluid replacement is present. At present, there is no standard diagnostic clinical or laboratory test for sepsis.[4] There is considerable variability in the reported levels of key cytokines involved in the pathogenesis of sepsis. Explanations for variability likely include different patient populations, different stages of sepsis (sepsis, severe sepsis, or septic shock), type of sample used (whole blood, plasma, or serum), and methods used to quantify cytokines.

The primary therapies for sepsis remain intravenous fluids, antibiotics, and vasopressors. Treatment includes general supportive care that can include ventilatory support with supplemental oxygen or mechanical ventilation and renal replacement therapy. It is postulated that targeting pro-inflammatory mediators would reduce mortality and morbidity of sepsis.[5 6] However, no specific treatment targeting these cytokines during sepsis has been proven to be effective in clinical trials.[7]

Current concepts of sepsis pathogenesis focus on an excessive host inflammatory response to infection that is mediated by elevated cytokine levels, also known as "Cytokine storm" or "Cytokine release syndrome". The molecular pathway that leads to a "cytokine storm" is vague.[8] The term "cytokine storm" appears to be first coined in a paper on Graft-Versus-Host disease.[8 9] In the early 2000s, this term was used in articles on infectious diseases.[8 10 11] Pivotal mediators of a cytokine storm include Tumour Necrosis Factor (TNF), interleukin-1 (IL-1)β, and interferon(IFN)γ. The cytokine storm is thought to cause organ malfunction and mortality that accompany sepsis.[8] Despite extensive study of cytokines in sepsis, the magnitude of cytokine levels and their effects on morbidity and mortality are poorly characterized. TNF, IL-1β, and IFNγ cytokines have been best characterized as pro-inflammatory and that are thought to drive the hyperinflammatory response to infection. This is underscored by the large number of clinical trials designed to specifically block these cytokines using selective inhibitors of TNF and IL-1β.[12 13] By our count, there have been at least nine late-stage

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clinical trials using selective TNF blocking drugs and three late-stage studies using IL-1β blockers. This does not include numerous studies (at least ten studies) where cytokines, in general, are inhibited using drugs like corticosteroids or endotoxin inhibitors (where inhibition will include TNF, IL-1β, and IFNγ). Given the large number of resources expended to block these pro-inflammatory molecules, it is remarkable that no one has published an overall assessment of the levels of these molecules that are present during sepsis. If you plan to block the activity of molecules (cytokines) that are thought to cause disease, it stands to reason you want to know how much of the molecules are present during the disease. If the putative cytokines that cause sepsis are present in unexpectedly low or surprisingly elevated concentrations, this is important information for understanding the cause of sepsis and for designing potential therapies. Moreover, there has been a spectacular failure of sepsis therapies designed to block cytokine activity. As stated by PE Marik ">100 phase 2 or 3 clinical trials of sepsis therapies have failed over the past three decades."[14] While not all of these trials directly targeted cytokine suppression, at least 47 clinical trials have targeted cytokine suppression in sepsis; all have failed. These observations underscore the importance of an improved understanding of what constitutes the goal of cytokine-blocking therapy; there is a need to know how many cytokines should be neutralized. This knowledge will inform both clinical and animal studies of sepsis and will help guide *in vitro* cytokine-blocking experiments.

If the cytokine storm concept of sepsis organ damage is correct, it is reasonable to assume that dysfunction of lungs (a requirement for mechanical ventilation) or kidneys (need for renal replacement therapy) associates with increased pro-inflammatory cytokine levels compared to sepsis patients without these organ failures. Inhibitors of TNF and IL-1β are used clinically to treat disease with hyper-inflammation as a component of pathogenesis (ex, rheumatoid arthritis). These anti-cytokine therapies carry "black box" warnings alerting prescribers that they increase the risk for and severity of infections. This documented association between cytokine inhibition and supervening infections suggests we should assess inflammatory cytokine levels for association with secondary infections (infections over and above the proven or presumed infection that caused sepsis). It is conjectured that lower cytokine levels in the circulation will correlate with an increased risk for secondary supervening infections. As well, it is expected that increased concentrations of the natural cytokine inhibitors TNFsRp55, TNFsRp75, or IL-1ra will associate with increased risk for secondary infections

No published or ongoing meta-analyses or systematic reviews evaluating key cytokine levels in sepsis patients and assessing the relationship between cytokine levels and sepsis outcomes were identified. The objective of this systematic review is to characterize levels of key cytokines in the circulation in patients with sepsis and assess the association between these levels and the morbidity and mortality outcomes related to sepsis.

Methods and Analysis

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This protocol adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) checklist. We will follow the PRISMA guidelines in the development of this systematic review.[15 16]

Protocol and Registration

Our systematic review protocol was registered with the International Prospective Register of Systematic Reviews (PROSPERO) on 5 July 2020 and was last updated on 21 December 2020 (registration number CRD42020179800).

Research questions

The systematic review will aim to answer the following questions:

1. What are the levels of key pro-inflammatory cytokines in sepsis?

2. How do these levels correlate with sepsis outcomes such as degree of sepsis (sepsis, severe sepsis, or septic shock), mortality, organ malfunction, and secondary infections?

Eligibility Criteria

Inclusion Criteria and Exclusion Criteria

Articles will be evaluated according to the criteria described below.

Study Design

This review will consider any design that reports relevant cytokine levels in patient cohorts. This review will consider longitudinal studies, prospective cohort studies, randomized and non-randomized clinical trials, case-control studies, and time-series studies. Case series and descriptive cross-sectional studies will also be included depending on the outcome measured. We will exclude case reports to minimize publication bias. Only studies published in English will be included. The studies must include cytokine concentrations in venous blood serum or plasma and express in mass units per volume. Quantified data depicted only in the graphical form will not be used. The assays used can be ELISA or immunologic/magnetic bead technologies. Studies that measured cytokines by a method other than ELISA or Immunologic/magnetic bead technologies will be excluded. We will include studies that defined sepsis according to the diagnostic criteria proposed by the American College of Chest Physicians and the Society of Critical Care Medicine in 1992 as the presence of systemic inflammatory response syndrome and infection. The 1992 definition was used instead of current definitions since most studies reporting cytokine levels utilized the 1992 definition. Participant studies subjects can be of any age, ethnicity, or sex. Sepsis studies criteria will include sepsis, severe sepsis, septic shock, purpura fulminans, or infection-associated DIC (a variant of organ malfunction involving coagulation system failure).

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Exposure:

The exposure of interest is the level of pro-inflammatory cytokines (TNF, IL-1β, and IFNγ) in patients with sepsis, severe sepsis or septic shock. The levels of TNF, IL-1β, or IFNɣ are assessed at presentation (t=0) or admission. We will assess the trend of cytokine levels after admission if such data are available. The cytokines are measured in venous blood (serum or plasma) using Enzyme-linked immunosorbent assay (ELISA) or Immunologic/magnetic bead technologies (Immunex™, others). We will extract the crude unadjusted mean and standard deviation from each cytokine in all studies. We will then calculate the standard error and log-transformed the values. We will enter the log-transformed mean and standard errors in a random-effect meta-analysis to combine the estimated log pooled values, which will be back-transformed for interpretation.

Outcomes:

The review will include the following outcome measures:

Primary outcome:

1. The mortality rate at 28 or 30-days.

Secondary outcomes:

1. Association with global organ dysfunction: SOFA score
2. Need for supplemental oxygen or mechanical ventilation.
3. Need for renal replacement therapy (RRT).
4. The emergence of secondary or supervening infections

Search Methods

Search strategy

The search strategy will include a combination of controlled vocabulary terms (specific to each database) and keyword terms. We will search for concepts including sepsis and cytokine levels of TNF, IL-1β, or IFNγ. Terms related to certain inclusion criteria, such as infection-associated DIC, are not included in the search strategy to maximize recall but will be factored in during full-text review. We did not include “cytokine” as a keyword because it is not specific, and we are more interested in the three pro-inflammatory cytokines (TNF, IL-1β, or IFN-ɣ) only. Appendix I lists a full search strategy for MEDLINE. The following electronic databases will be searched: Medline (via Ovid), Embase (via Elsevier), Cochrane Library (via Wiley, including Cochrane Central Register of Controlled Trials), and Web of Science Core Collection (via Clarivate Analytics, including Science Citation Index Expanded and Social Sciences Citation Index). The reference list of all studies selected for critical appraisal will be screened for additional studies. There will be no restrictions or limits on the age or sex of subjects. The language will be restricted to English. Filters will be used to limit results to human studies (Appendix I, Table 1).

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Study selection

After the search, all identified studies will be uploaded to Endnote X9 (Clarivate Analytics, PA, USA) and Covidence.[17] We will remove duplicate studies. There will be two levels of screening. First, two reviewers (OS and AM) will independently screen the titles and abstracts of studies for eligibility and data extraction. Next, the two reviewers will then independently evaluate the full-text articles for inclusion/exclusion. Should discrepancies arise, the two reviewers will resolve the disagreements by discussion. Should that fail to be resolved, a third reviewer (AHM) will reconcile disagreements between the two reviewers through expert advice. Through Covidence, we will generate a PRISMA document with the number of results we started with, the number excluded during title/abstract screening, and the number excluded during full-text assessments, along with reasons for exclusion.. We will also manually include pre-selected studies that we consider important, and that were not captured during the search. We will include the final number in the final report.

Data Extraction

Data will be extracted using a standardized extraction form. Extracted data will consist of: type of study, number of participants, study methods, the technology used to measure the three pro-inflammatory cytokines (TNF, IL-1β, and IFNγ), and outcomes of significance as described above. We will also extract clinical factors in the patient cohorts examined. Such factors might include age, comorbidities, gender, race, BMI, HIV infection, other forms of immunosuppression (chemotherapy, radiation, leukemia, and lymphoma), type of sepsis (bacterial, viral, fungal, etc.), sepsis severity (sepsis, severe sepsis, and septic shock), the site of origin of the infection (blood, urine, skin and soft tissue, respiratory, other). The level of organ dysfunction in the cohorts will be assessed by recording the SOFA score during the first 24 hours after admission.

Study data will be collected and managed using REDCap electronic data capture tools hosted at the University of Colorado Denver. Additional relevant information that arises during data extraction will be recorded, and this will be reported in the final systematic review manuscript. Two independent persons (OS and AM) will be involved with data extraction. Disagreements will be resolved by the assessment of a third reviewer (AHM). All extraction forms, tools, and meta-analyses will be pilot tested in two studies and subsequently modified to ensure the extraction of relevant data and assess the actual feasibility of our research question. If any data are missing from a study or should uncertainties arise, we will attempt to contact the study authors to obtain the missing data or clarify any ambiguity.

Quality Assessment

Two reviewers (OS and AM) will independently evaluate the included studies for risk of bias, and any disagreements will be resolved by a third reviewer (AHM). The methodological quality of included studies will be assessed independently by the same two reviewers according to the Quality in Prognosis Studies (QUIPS) tool.[18] The QUIPS tool considered 6 domains of bias: study participation, study attrition, prognostic factor measurement, outcome measurement, study confounding, and statistical analysis and reporting. Each domain is comprised of a list of items that enable assessment

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of the domain as low, moderate, or high risk. If the majority of criteria were met, a "++" rating was given. If some of the criteria were met, a "+" rating was given. If most of the criteria is not met, studies were rated as low-quality, and they were excluded.

Risk of Bias

We will produce a funnel plot to assess for publication bias. Studies with high precision will be plotted closer to the average, and those with low precision will spread on either side of the average, producing a funnel-shaped distribution. The more asymmetric the shape is, the more profound the amount of bias in the meta-analysis is.[19]

Data Synthesis

Studies will, where possible, be pooled in a statistical meta-analysis. All data will be presented and synthesized through tables' summaries, figures, and charts, where appropriate. Quantitative data will be pooled with statistical meta-analysis. Effect sizes will be expressed as odds ratios or risk ratios for dichotomous data and weighted mean differences (for continuous data), and their 95% confidence intervals will be calculated. Will report study level variables, and they will be evaluated by their mean or proportion. Study level heterogeneity will be assessed using the I-square test. We plan to do a fixed or random effect meta-analysis method depending on the studies' heterogeneity and sample size.

Exploratory subgroup analyses will be performed, where possible, based on the following:

- Studies mean age as continuous or dichotomized using the mean
- Studies primary cytokine measurement technique (ELISA vs. immunologic)
- Year of the study publication, as continuous
- Type of studies (prospective, case-control, cross-sectional, or randomized clinical trials)
- The proportion of studies participants per immune function status (Immunocompetent versus immunocompromised) or comorbidities (congestive heart failure or malignancy; yes or no)
- Cohort sample size, as continuous or dichotomized using the mean
- Studies proportion of gender; as continuous or dichotomized using the mean
- Studies proportions of the type of sepsis source included
- Studies proportions of sepsis severity
- Studies with the inclusion of a specific sepsis source (respiratory vs. not; abdominal sepsis vs. not, urinary sepsis vs. not, etc.)
- Studies mean SOFA scores; as continuous or dichotomized using the mean

Continuous variables extracted means will be checked for linearity using a two-way scatter plot against log-transformed cytokine mean levels and a coefficient correlation using meta-regression.

Assessing certainty in the findings

We will follow the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) system for quantifying the quality of evidence. A Summary of Findings (SoF) will be created using

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GRADEPro GDT as feasible. All primary outcomes will be included in the SoF. The SoF will present the following information, where appropriate: a ranking of the quality of the evidence based on the risk of bias, directness, heterogeneity, precision, and risk of publication bias of the review results. Summary results from meta-analyses will be reported as summary point estimates and interval estimates. The meta-analysis forest plots will be presented as appropriate. If used, results of a funnel plot for publication bias or risk of publication bias will be presented. If meta-analysis is not performed, a narrative summary in words and tables will be included.

Ethics and Dissemination

Ethical approval is not required for this systematic review since it aims to collect data from existing literature. As such, there are no privacy concerns. Findings will be disseminated in the form of peer-reviewed publications and abstract presentations at research conferences.

In conclusion, our review will evaluate the available evidence for key cytokine levels and their association with sepsis outcomes using the GRADE approach.[20] We believe that this work will address an important gap by reporting on a gold standard reference for cytokine levels in sepsis. This work will also address the concept of “cytokine storm” in sepsis. Our findings may guide future sepsis management strategies, such as cytokine-blocking therapies.

Patient and Public Involvement

Patients or the public are not involved in the design, conduct, reporting, or dissemination plans of this research work.

Author Contributions

AHM designed the systematic review protocol with the help of the librarian KD. AAG and LS prepared the first draft of the manuscript. OS, AM, SS, KD, SS, CFP, AHM, and LS reviewed and revised the first draft. All authors read and approved the manuscript.

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Role of the Funder/Sponsor

The funder had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Conflicts of Interest

The authors declare no conflict of interest.

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Patient consent for publication

Not required

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References

1. Rudd KE, Johnson SC, Agesa KM, et al. Global, regional, and national sepsis incidence and mortality, 1990–2017: analysis for the Global Burden of Disease Study. *The Lancet* 2020;395(10219):200-11.
2. Torio C, Moore B. National inpatient hospital costs: the most expensive conditions by payer, 2013. HCUP statistical brief no. 204. Rockville, MD: Agency for Healthcare Research and Quality; May 2016, 2017.
3. Bone RC, Sibbald WJ, Sprung CL. The ACCP-SCCM consensus conference on sepsis and organ failure. *Chest* 1992;101(6):1481-83.
4. Singer M, Deutschman CS, Seymour CW, et al. The third international consensus definitions for sepsis and septic shock (Sepsis-3). *Jama* 2016;315(8):801-10.
5. Jekarl DW, Kim JY, Ha JH, et al. Diagnosis and Prognosis of Sepsis Based on Use of Cytokines, Chemokines, and Growth Factors. *Disease markers* 2019;2019
6. Huang M, Cai S, Su J. The Pathogenesis of Sepsis and Potential Therapeutic Targets. *International journal of molecular sciences* 2019;20(21):5376.
7. Cytokine storm and sepsis disease pathogenesis. Seminars in immunopathology; 2017. Springer.
8. Tisoncik JR, Korth MJ, Simmons CP, et al. Into the eye of the cytokine storm. *Microbiol Mol Biol Rev* 2012;76(1):16-32.
9. Cytokine storm of graft-versus-host disease: a critical effector role for interleukin-1. Transplantation proceedings; 1993.

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10. Imashuku S. Clinical features and treatment strategies of Epstein-Barr virus-associated hemophagocytic lymphohistiocytosis. *Crit Rev Oncol Hematol* 2002;44(3):259-72. doi: 10.1016/s1040-8428(02)00117-8
11. Jahrling PB, Hensley LE, Martinez MJ, et al. Exploring the potential of variola virus infection of cynomolgus macaques as a model for human smallpox. *Proc Natl Acad Sci U S A* 2004;101(42):15196-200. doi: 10.1073/pnas.0405954101
12. Abraham E, Anzueto A, Gutierrez G, et al. Double-blind randomised controlled trial of monoclonal antibody to human tumour necrosis factor in treatment of septic shock. *The Lancet* 1998;351(9107):929-33.
13. Bone RC. Why sepsis trials fail. *Jama* 1996;276(7):565-66.
14. Marik PE, Khangoora V, Rivera R, et al. Hydrocortisone, vitamin C, and thiamine for the treatment of severe sepsis and septic shock: a retrospective before-after study. *Chest* 2017;151(6):1229-38.
15. Moher D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic reviews* 2015;4(1):1.
16. Moher D, Altman DG, Liberati A, et al. PRISMA statement. *Epidemiology* 2011;22(1):128.
17. Innovation V. Covidence systematic review software. *Melbourne, Australia* 2017
18. Hayden JA, van der Windt DA, Cartwright JL, et al. Assessing bias in studies of prognostic factors. *Annals of internal medicine* 2013;158(4):280-86.
19. Higgins JP, Thomas J, Chandler J, et al. Cochrane handbook for systematic reviews of interventions: John Wiley & Sons 2019.
20. Guyatt G, Oxman AD, Akl EA, et al. GRADE guidelines: 1. Introduction—GRADE evidence profiles and summary of findings tables. *Journal of clinical epidemiology* 2011;64(4):383-94.

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Appendix 1. Search strategy

MEDLINE (via Ovid MEDLINE® ALL, 1946 to present)	
Search date = 5/1/2020	
1	Sepsis/ or Shock, Septic/ or Purpura Fulminans/ or (sepsis or septic or purpura fulminans).tw,kf.
2	Interferon-gamma/ or (((interferon or IFN) adj1 (gamma or y)) or IFNgamma or IFNy).tw,kf.
3	Tumor Necrosis Factor-alpha/ or (((tumor necrosis factor or TNF) adj1 (alpha or a)) or TNFalpha or TNFa or ((tumor necrosis factor or TNF) adj1 soluble adj1 receptor adj2 ("1" or I or "2" or II)) or TNFsRp55 or TNFsRp75).tw,kf.
4	Interleukin-1beta/ or (((interleukin or IL) adj1 (1beta or 1b or 1-beta or 1-b)) or IL1beta or IL1b).tw,kf. or Interleukin 1 Receptor Antagonist Protein/ or (((interleukin or IL) adj1 "1" adj1 receptor antagonist) or IL-1ra or IL1ra).tw,kf.
5	or/2-4
6	Case-Control Studies/ or exp Clinical Trial/ or Clinical Trials as Topic/ or Controlled Clinical Trial/ or Cohort Studies/ or Cross-Sectional Studies/ or Longitudinal Studies/ or Prospective Studies/ or Randomized Controlled Trial/ or (case series or case-control or clinical trial or cohort study or cross-sectional or longitudinal study or prospective cohort* or (random* adj1 (clinical or control*) adj1 trial)).tw,kf. or randomized.ab. or placebo.ab. or randomly.ab. or trial.ti.
7	1 and 5 and 6
8	limit 7 to english language
9	exp animals/ not exp humans/
10	8 not 9
11	limit 10 to yr="1985 -Current"

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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1

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Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a (This is not an update of a previous systematic review)
Registration			
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Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	10
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a (not an amendement)
Support			
Sources	#5a	Indicate sources of financial or other support for the review	10

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Sponsor	#5b	Provide name for the review funder and / or sponsor	n/a (No funding)
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	n/a (No funding)
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	3,4,5
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7

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Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	13
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7-9
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	8
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7

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Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	9
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	9-10
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	n/a
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	9-10

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Notes:

- 1b: n/a (This is not an update of a previous systematic review)
- 4: n/a (not an amendement)
- 5b: n/a (No funding)
- 5c: n/a (No funding) The PRISMA-P checklist is distributed under the terms of the Creative Commons Attribution License CC-BY 4.0. This checklist was completed on 26. December 2020 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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Author and year of publication			
Study identifier Reviewer			
Biases	**Issues to consider for judging overall rating of "Risk of bias"**	**Study Methods & Comments**	**Rating of reporting**
Instructions to assess the risk of each potential bias:	These issues will guide your thinking and judgment about the overall risk of bias within each of the 6 domains. Some 'issues' may not be relevant to the specific study or the review research question. These issues are taken together to inform the overall judgment of potential bias for each of the 6 domains.	Provide comments or text exerpts in the white boxes below, as necessary, to facilitate the consensus process that will follow.	Click on each of the blue cells and choose from the drop down menu to rate the adequacy of reporting as yes, partial, no or unsure.
1. Study Participation	**Goal: To judge the risk of selection bias (likelihood that relationship between *PF* and *outcome* is different for participants and eligible**		
Source of target population	The source population or population of interest is adequately described for key characteristics (LIST).		
Method used to identify population	The sampling frame and recruitment are adequately described, including methods to identify the sample sufficient to limit potential bias (number and type used, e.g., referral patterns in health care)		
Recruitment period	Period of recruitment is adequately described		
Place of recruitment	Place of recruitment (setting and geographic location) are adequately described		
Inclusion and exclusion criteria	Inclusion and exclusion criteria are adequately described (e.g., including explicit diagnostic criteria or "zero time" description).		
Adequate study participation	There is adequate participation in the study by eligible individuals		
Baseline characteristics	The baseline study sample (i.e., individuals entering the study) is adequately described for key characteristics (LIST).		
Summary Study participation	**The study sample represents the population of interest on key characteristics, sufficient to limit potential bias of the observed relationship between PF and outcome.**		
2. Study Attrition	**Goal: To judge the risk of attrition bias (likelihood that relationship between *PF* and *outcome* are different for completing and non-completing participants).**		
Proportion of baseline sample available for analysis	Response rate (i.e., proportion of study sample completing the study and providing outcome data) is adequate.		
Attempts to collect information on participants who dropped out	Attempts to collect information on participants who dropped out of the study are described.		
Reasons and potential impact of subjects lost to follow-up	Reasons for loss to follow-up are provided.		
Outcome and prognostic factor information on those lost to follow-up	Participants lost to follow-up are adequately described for key characteristics (LIST).		
	There are no important differences between key characteristics (LIST) and outcomes in participants who completed the study and those who did not.		

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Study Attrition Summary	Loss to follow-up (from baseline sample to study population analyzed) is not associated with key characteristics (i.e., the study data adequately represent the sample) sufficient to limit potential bias to the observed relationship between PF and outcome.		

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3. Prognostic Factor Measurement	**Goal: To judge the risk of measurement bias related to how PF was measured (differential measurement of PF related to the level of outcome).**		
Definition of the PF	A clear definition or description of 'PF' is provided (e.g., including dose, level, duration of exposure, and clear specification of the method of measurement).		
Valid and Reliable Measurement of PF	Method of PF measurement is adequately valid and reliable to limit misclassification bias (e.g., may include relevant outside sources of information on measurement properties, also characteristics, such as blind measurement and limited reliance on recall).		
	Continuous variables are reported or appropriate cut-points (i.e., not data-dependent) are used.		
Method and Setting of PF Measurement	The method and setting of measurement of PF is the same for all study participants.		
Proportion of data on PF available for analysis	Adequate proportion of the study sample has complete data for PF variable.		
Method used for missing data	Appropriate methods of imputation are used for missing 'PF' data.		
PF Measurement Summary	***PF* is adequately measured in study participants to sufficiently limit potential bias.**		

4. Outcome Measurement	**Goal: To judge the risk of bias related to the measurement of outcome (differential measurement of outcome related to the baseline level of PF).**		
Definition of the Outcome	A clear definition of outcome is provided, including duration of follow-up and level and extent of the outcome construct.		
Valid and Reliable Measurement of Outcome	The method of outcome measurement used is adequately valid and reliable to limit misclassification bias (e.g., may include relevant outside sources of information on measurement properties, also characteristics, such as blind measurement and confirmation of outcome with valid and reliable test).		
Method and Setting of Outcome Measurement	The method and setting of outcome measurement is the same for all study participants.		
Outcome Measurement Summary	***Outcome of interest* is adequately measured in study participants to sufficiently limit potential bias.**		

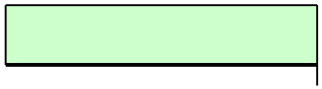
5. Study Confounding	**Goal: To judge the risk of bias due to confounding (i.e. the effect of PF is distorted by another factor that is related to PF and outcome).**		
Important Confounders Measured	All important confounders, including treatments (key variables in conceptual model: LIST), are measured.		
Definition of the confounding factor	Clear definitions of the important confounders measured are provided (e.g., including dose, level, and duration of exposures).		
Valid and Reliable Measurement of Confounders	Measurement of all important confounders is adequately valid and reliable (e.g., may include relevant outside sources of information on measurement properties, also characteristics, such as blind measurement and limited reliance on recall).		
Method and Setting of Confounding Measurement	The method and setting of confounding measurement are the same for all study participants.		
Method used for missing data	Appropriate methods are used if imputation is used for missing confounder data.		
Appropriate Accounting for Confounding	Important potential confounders are accounted for in the study design (e.g., matching for key variables, stratification, or initial assembly of comparable groups).		
	Important potential confounders are accounted for in the analysis (i.e., appropriate adjustment).		
Study Confounding Summary	**Important potential confounders are appropriately accounted for, limiting potential bias with respect to the relationship between *PF* and *outcome*.**		

6. Statistical Analysis and Reporting	**Goal: To judge the risk of bias related to the statistical analysis and presentation of results.**		
Presentation of analytical strategy	There is sufficient presentation of data to assess the adequacy of the analysis.		
Model development strategy	The strategy for model building (i.e., inclusion of variables in the statistical model) is appropriate and is based on a conceptual framework or model.		
	The selected statistical model is adequate for the design of the study.		
Reporting of results	There is no selective reporting of results.		
Statistical Analysis and Presentation Summary	**The statistical analysis is appropriate for the design of the study, limiting potential for presentation of invalid or spurious results.**		

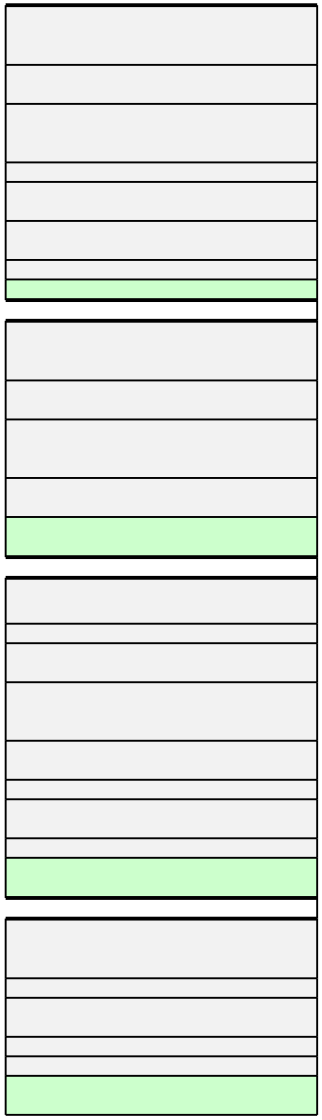
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Rating of "Risk of bias"
Click on the green cells; choose from the drop-down menu to rate potential risk of bias for each of the 6 domains as High, Moderate, or Low considering all relevant issues

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